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# Improving Attention-Enhanced Reservoir Computing with Intrinsic Plasticity

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*Dedicated to  
my family and my friends.*



## Abstract

While the Transformer architecture has dominated the machine learning field this decade, there is a growing search for more efficient architectures during both training and inference. Reservoir computing, while remaining a strong class of models for time series prediction, struggles with discrete data like text. Attention-enhanced reservoir computing (AERC) tries to overcome this limitation using attention only on a fixed size reservoir, avoiding a growing KV cache. In this paper, we integrate Intrinsic Plasticity to the fixed reservoir to improve the AERC architecture and test it on a wider set of tasks. We find TODO



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# Chapter 1

## Introduction

In recent years there has been a growing search for efficient, subquadratic alternatives to Transformer, to make training and inference cheaper. The main approaches use a lighter variant of attention, such as Native Sparse Attention [11], or a linear architecture, such as Mamba[4] or RWKV[9].

One architecture that has attracted interest for sequence modeling within the class of recurrent neural networks is the Reservoir Computing (RC) model[5], which uses a fixed recurrent reservoir for non-linear temporal expansion of the inputs, and a linear layer to extract meaning for the high-dimensional data. Reservoir Computing models, while showing state of the art results in time series applications, struggle with discrete data, and have been therefore ignored for text autoregression. The promising aspects of RC are in the fixed, recurrent, non-linear reservoir and the trainable linear output layer, which can be trained using Ridge Regression. In addition, the use of a physical reservoir makes this process even more efficient, while increasing the capabilities of the model[1].

Building upon the RC model there have been studies to expand its capabilities using multiple reservoirs [2], replacing the reservoir with a direct nonlinear feature-expansion model[3], enforcing the Echo State Property for the reservoir connection matrix[5], using leaky neurons[6], feedback connections[8] and Intrinsic Plasticity(IP)[10]. While these approaches have improved the base RC model, they still struggle with discrete data.

For this reason, an Attention-Enhanced Reservoir Computing (AERC) model has been proposed [7], using dynamically generated attention weights, greatly improving the performance of the model with discrete data while still leveraging the cheap recurrence of the fixed reservoir.

In this thesis we: - integrate IP in the AERC model



## Chapter 2

# Literature Review

Text for the literature review.



## Chapter 3

# Methodology

Text for the methodology.



## Chapter 4

# Results

Text for the results.





## Chapter 5

# Discussion

Text for the discussion.



## Chapter 6

# Conclusion

Text for the conclusion.



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